



**AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)**

**Department of Computer Science  
Faculty of Science & Technology (FST)**

**Fall 2024-25**

**Section: H, Group: 01**

**PROJECT TITLE**

**Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online  
Learning Environment**

**Supervised By TONNY SHEKHA KAR**

**Submitted By**

**Group Members**

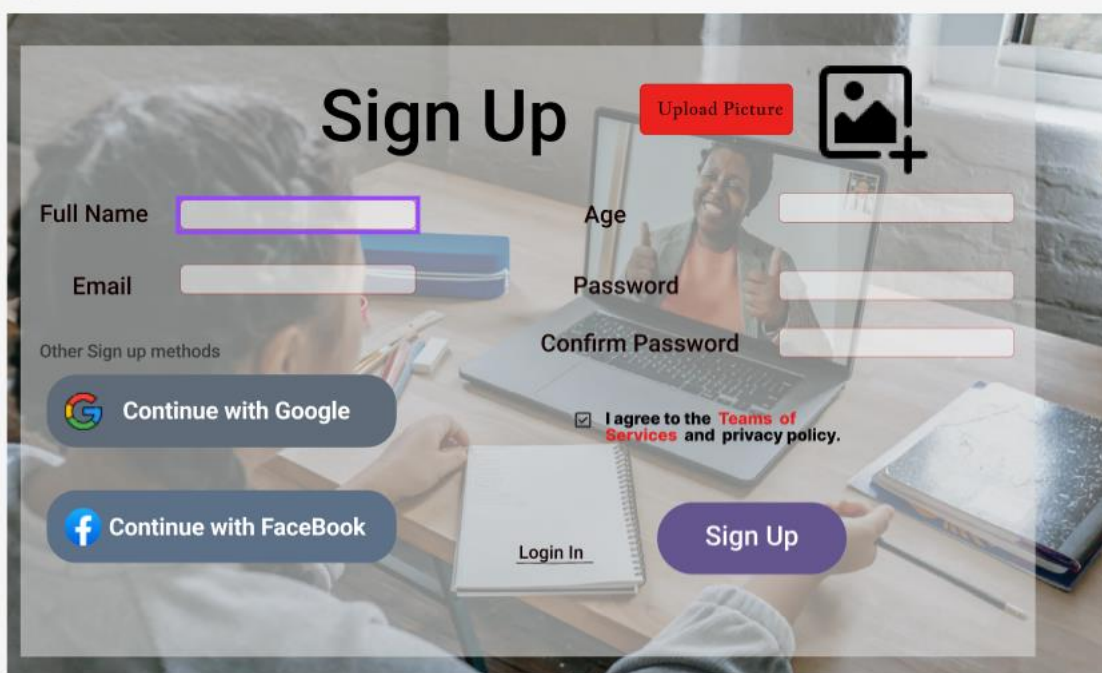
| SN | Name                  | ID         |
|----|-----------------------|------------|
| 1  | AFNAN BIN ISLAM NAHIN | 22-49350-3 |
| 2  | AMAN ULLAH            | 22-49345-3 |
| 3  | HRITTIK BHATTACHARJEE | 21-44650-1 |
| 4  | MD. HACHIBUR RAHMAN   | 22-46426-1 |
| 5  | WALID BIN KHALED      | 22-46101-1 |

## Lab 1 Final: UI Design

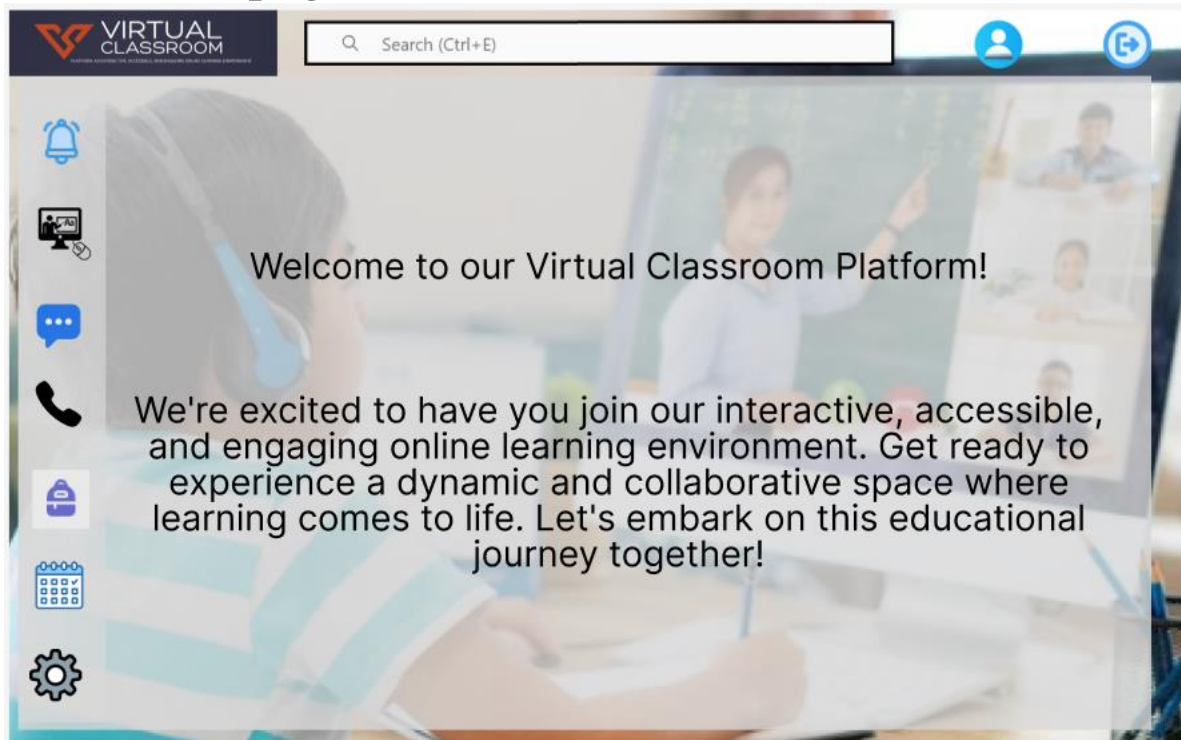
### Login page



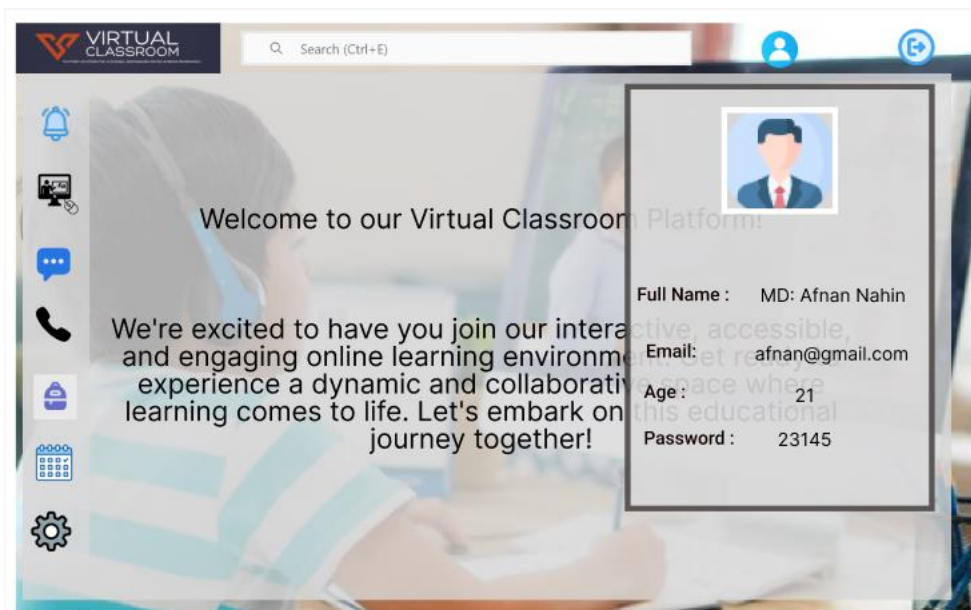
### SignUp page



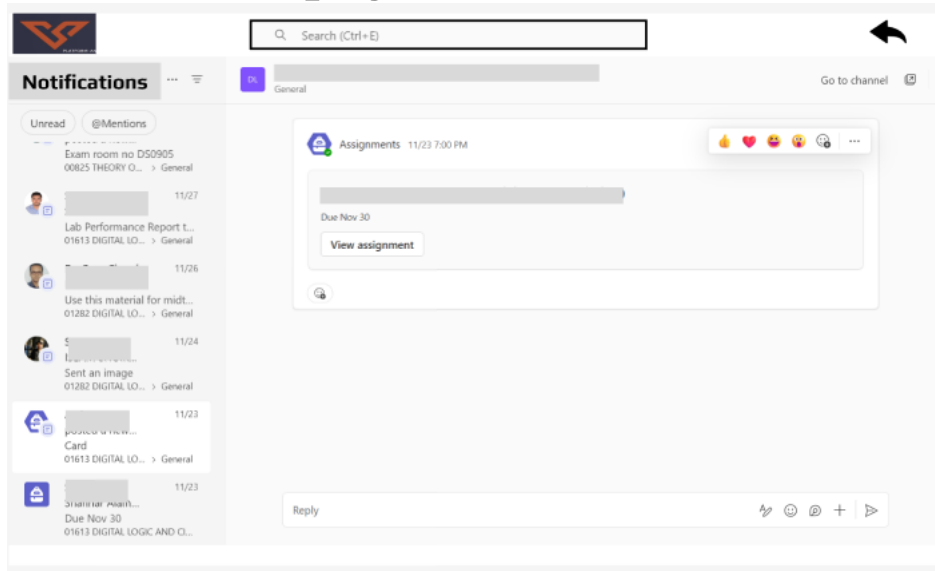
## Dashboard page



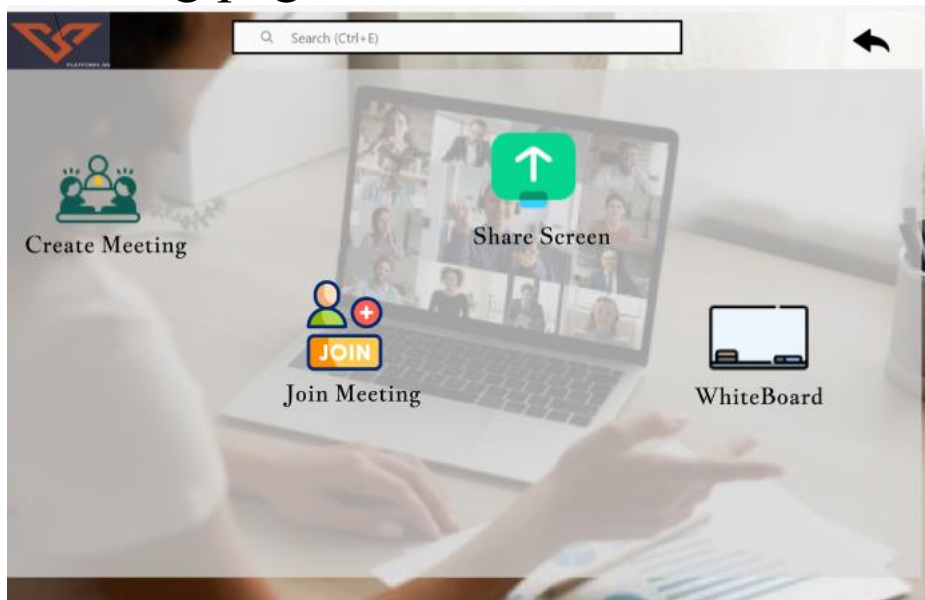
## Profile page



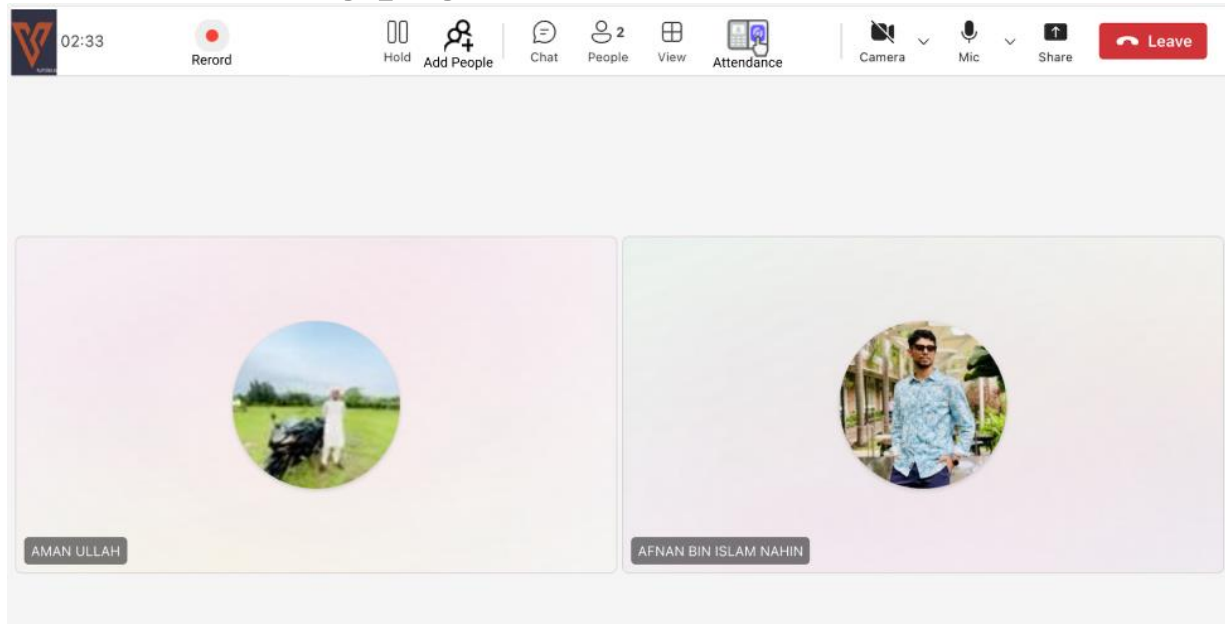
# Notification page



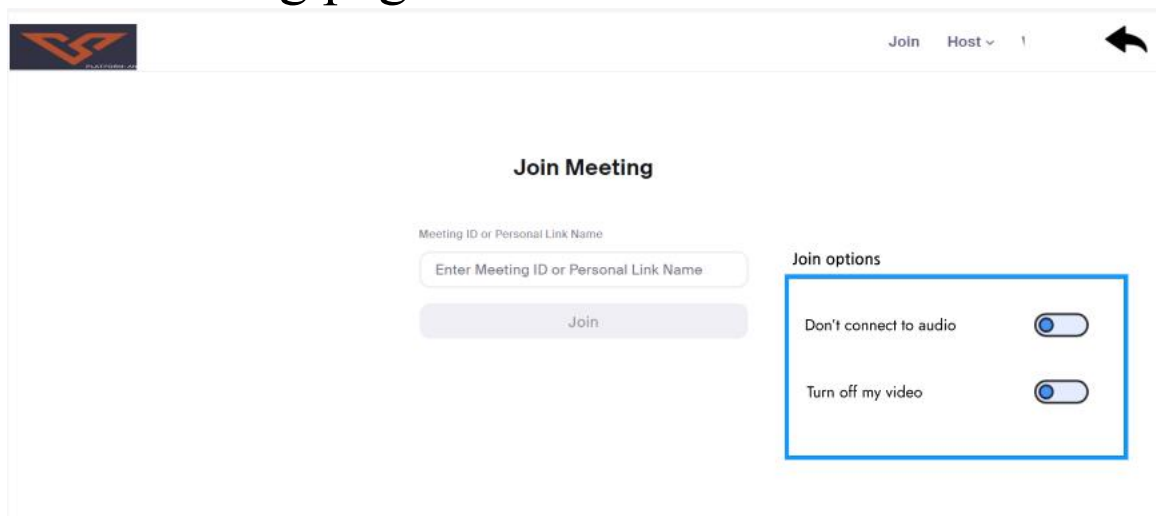
# Meeting page and all buttons



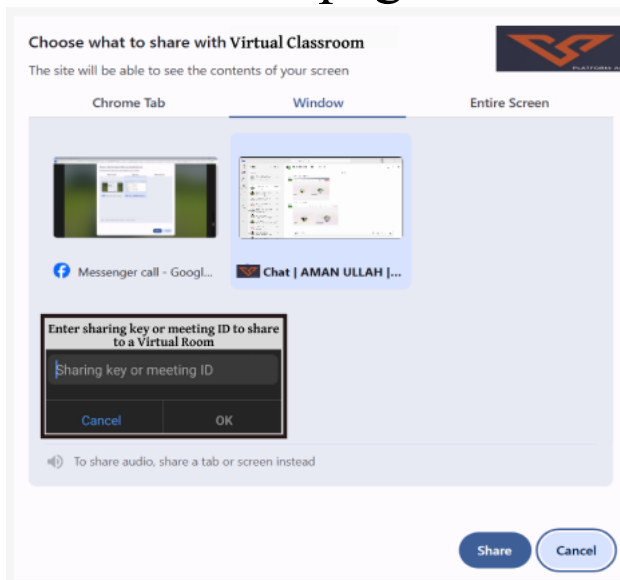
## Create meeting page



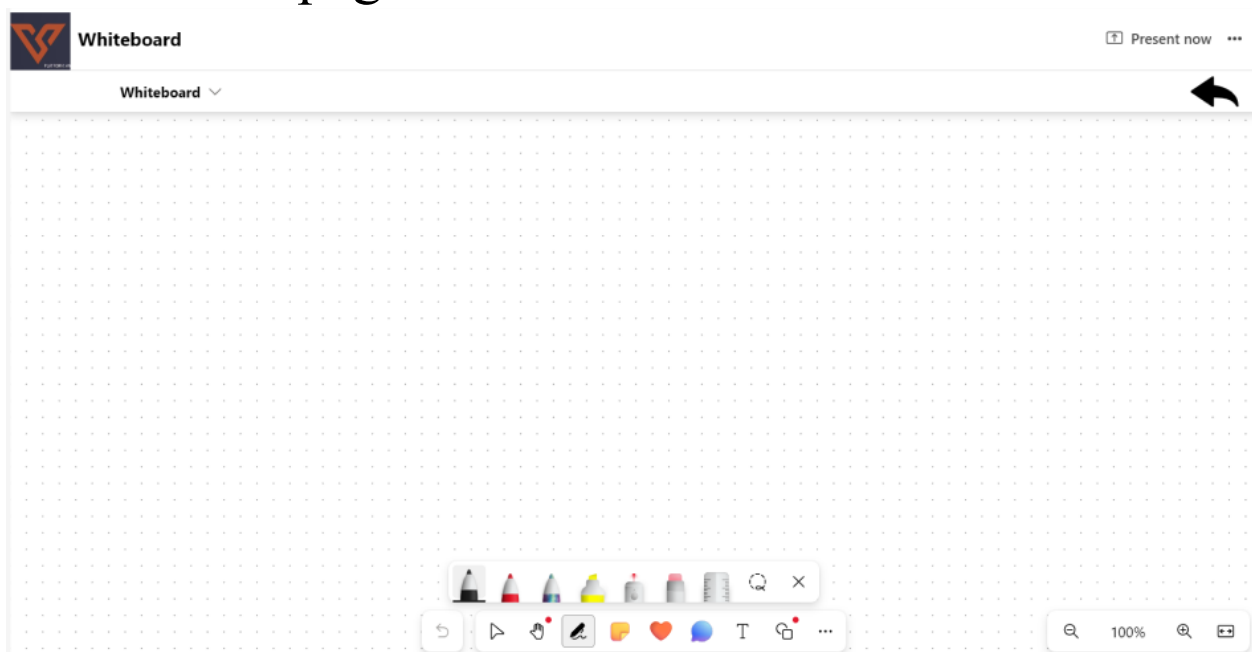
## Join meeting page



## Screen share page



## Whiteboard page



# Chat Page



Chat

10/10  
Pls go to Engineering - Comput...

10/6  
You: <https://www.facebook.com...>

10/6  
You: Sir, I request you to make ...

10/6  
...

9/16  
...

9/10  
bro

9/3  
private void ConfigureDataGrid...

8/21  
brother device er lab reportgu...

8/18  
...re request de you are up

8/14  
Not really I expect next week

7/2  
Thanks bro

6/28  
Multi-variate signal pr...  
Meeting ended

Current price: 100 ( $150 - 50 = 100$ )

Buy Signal generated

look at example below



1.

Also we need to make portfolio

<https://vmonissette.github.io/bt/examples.html>

# Call page



Calls

Personal

Type a name

MM

R...

AA

A...

Call

History

All

Outgoing

Thursday

Outgoing

2m 4s

11/30/2024

Outgoing

11/14/2024

Outgoing

16s

11/13/2024

Outgoing

6/25/2024

Outgoing

6/25/2024

Incoming from ...

28m 18s

6/15/2024

Incoming from ...

5/16/2024

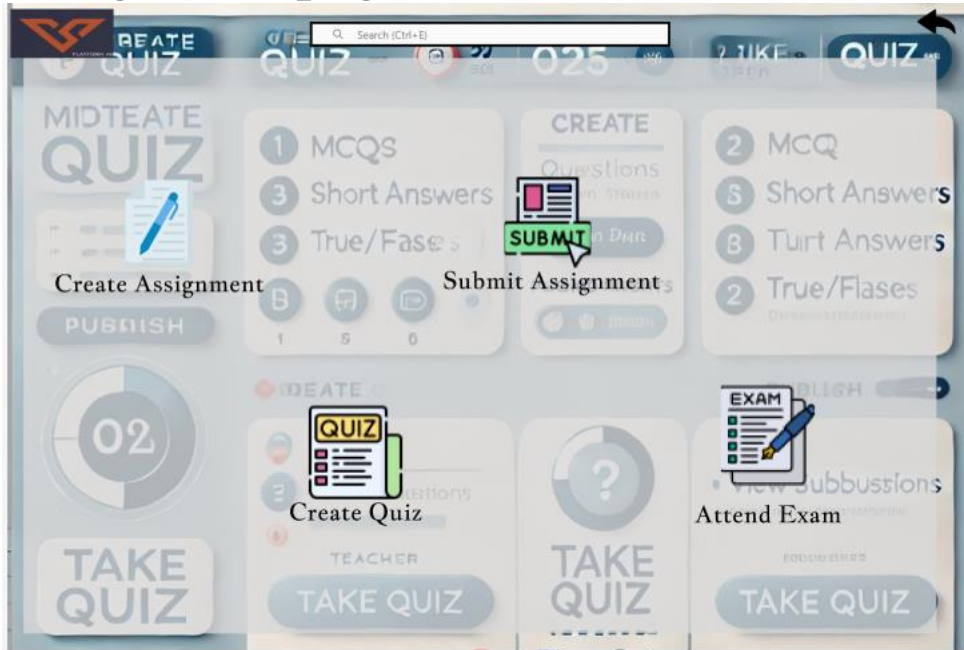
Outgoing

5/2/2024

Forwarded to voicemail

5/1/2024

## Assignment page



## Create assignment page

The screenshot shows the 'Create assignment' form. It includes a title field 'Add Assignment Title', a 'Points' field, and an 'Instructions' field. Below these is a 'My work' section with 'Attach' and 'New' buttons. A 'Share' button is at the bottom left. A 'Select date' dialog box is open, showing 'Enter deadline' and a date input field with the placeholder 'mm/dd/yyyy'. The dialog has 'Cancel' and 'OK' buttons.

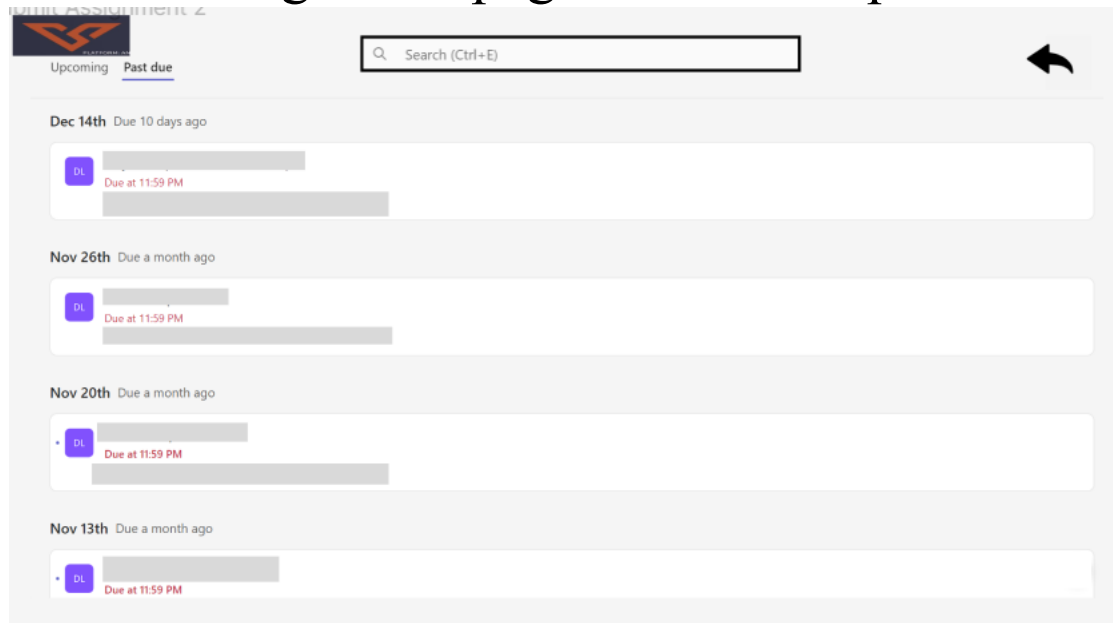
## Create quiz page

The screenshot shows a web interface for creating a quiz. At the top left is a logo with the text 'PLATFORM'. To the right of the logo are icons for bookmark, chat, eye, undo, redo, and link. A blue 'Share' button is in the top right corner. Below the logo, the word 'Questions' is centered. The main area has a purple header with 'Add Title' and 'Form description'. Below this is a question editor. The question title is 'Untitled Question'. To the right of the title is a dropdown menu set to 'Multiple choice'. Below the title are two radio button options: 'Option 1' and 'Add option or Add "Other"'. At the bottom right of the question editor are icons for copy, delete, and a 'Required' toggle switch. On the right side of the question editor is a vertical toolbar with icons for adding questions, deleting, undo, redo, and other actions.

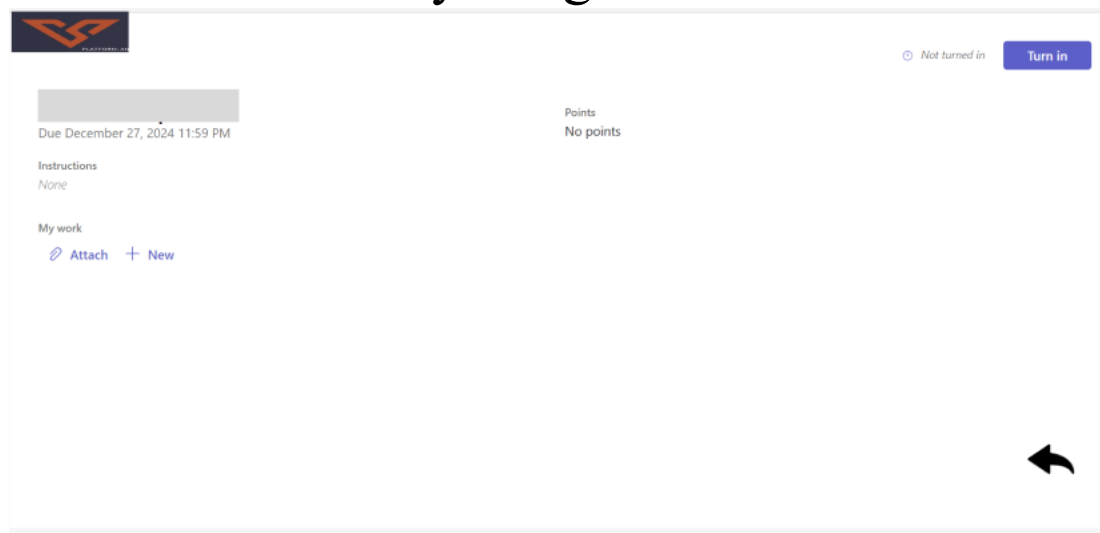
## Submit assignment page

The screenshot shows a web interface for submitting assignments. At the top left is a logo with the text 'PLATFORM'. To the right of the logo is a search bar with the text 'Search (Ctrl+E)'. Below the logo, the words 'Upcoming' and 'Past due' are displayed. Below this is a section for 'Dec 27th Friday'. This section contains two assignment cards. Each card has a purple 'DL' icon, a title bar, and a progress bar. Below the 'Dec 27th Friday' section is a section for 'Jan 15, 2025 Wednesday'. This section also contains two assignment cards, each with a purple 'DL' icon, a title bar, and a progress bar. A black arrow icon is in the top right corner.

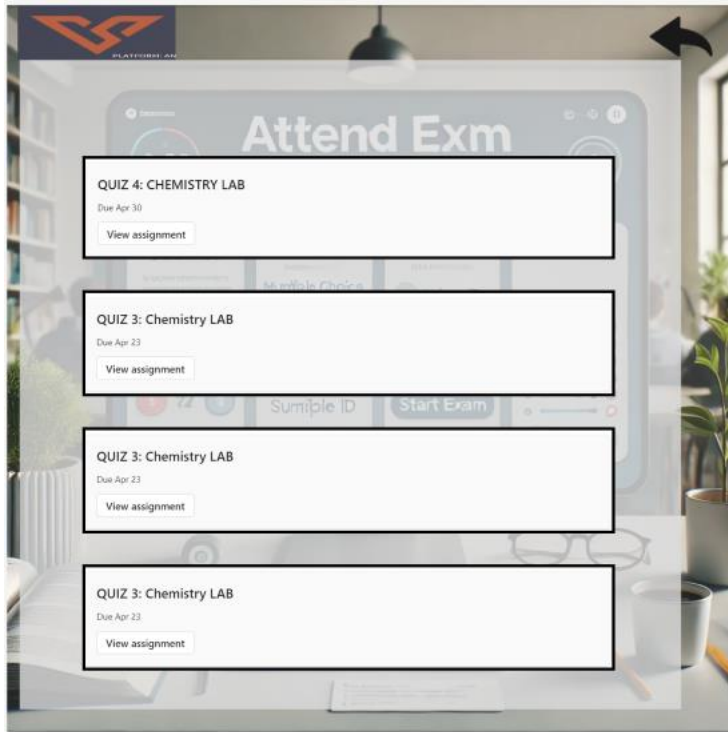
# Submit assignment page when click past due



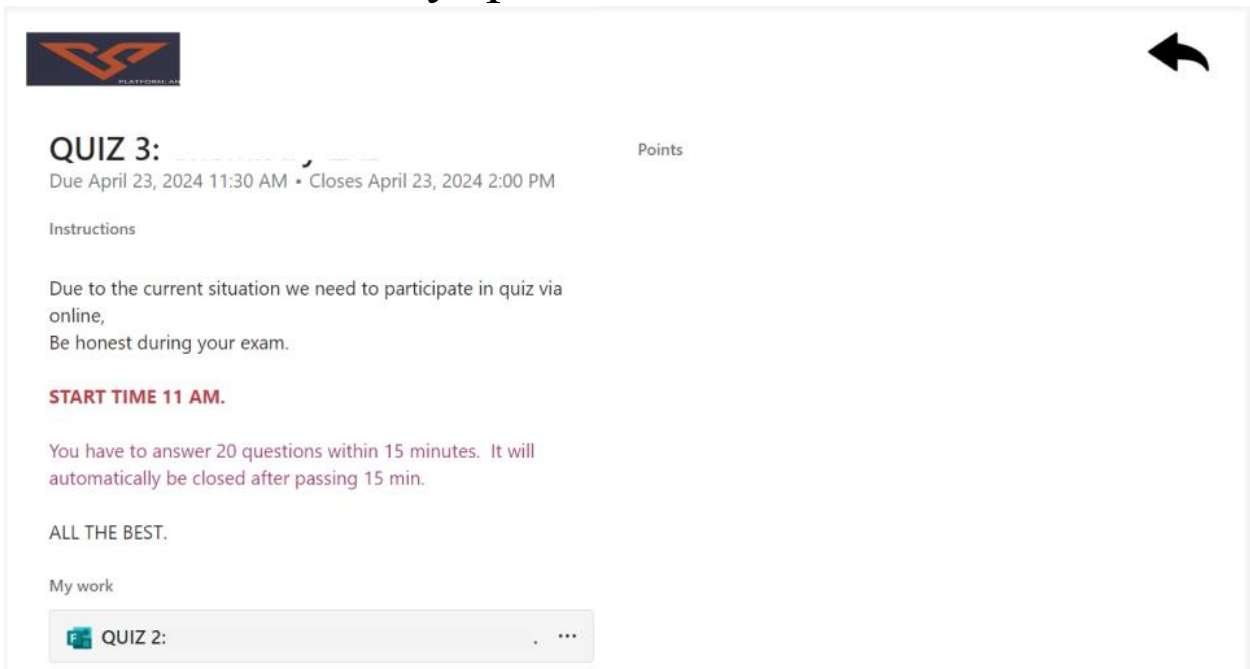
# When we click any assignment



## Attend exam page



## When we click any quiz



# Exam page

 PLATFORM AG

38:24

^

☐ 20,000 burgers

☐ 4,800 burgers

33. Accumulated depreciation balance cannot exceed the \_\_\_\_\_ (1 Point) 

☐ useful life

☐ net book value

☐ residual salvage value

☒ depreciable cost

34. Which one of the following assets will NEVER be depreciated? (1 Point) 

☐ Air Conditioner

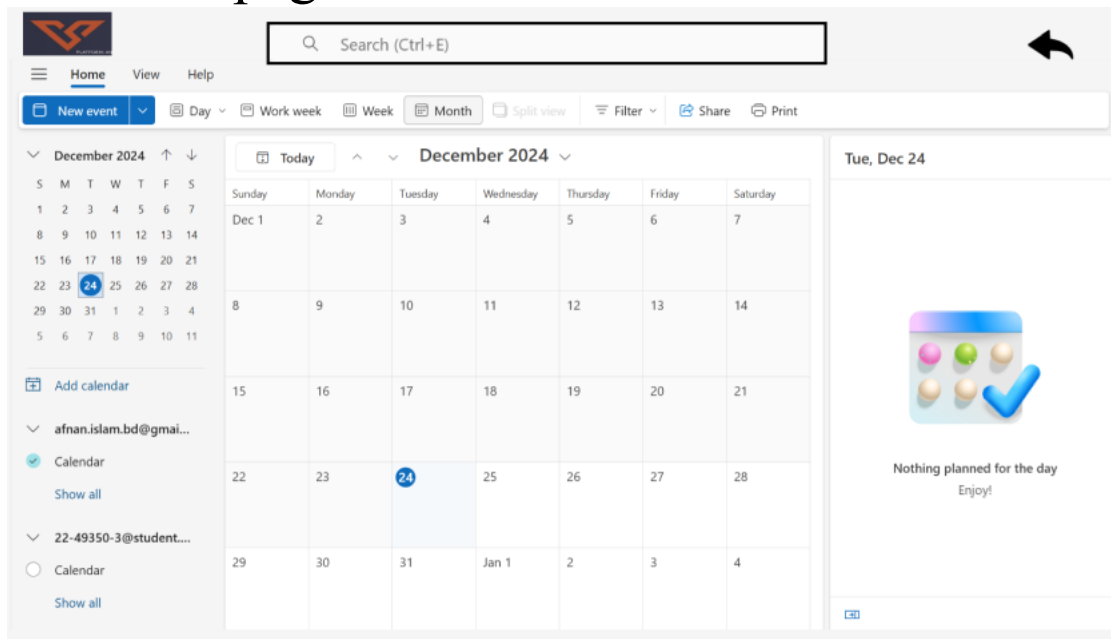
☐ Factory building

☐ Parking Lot

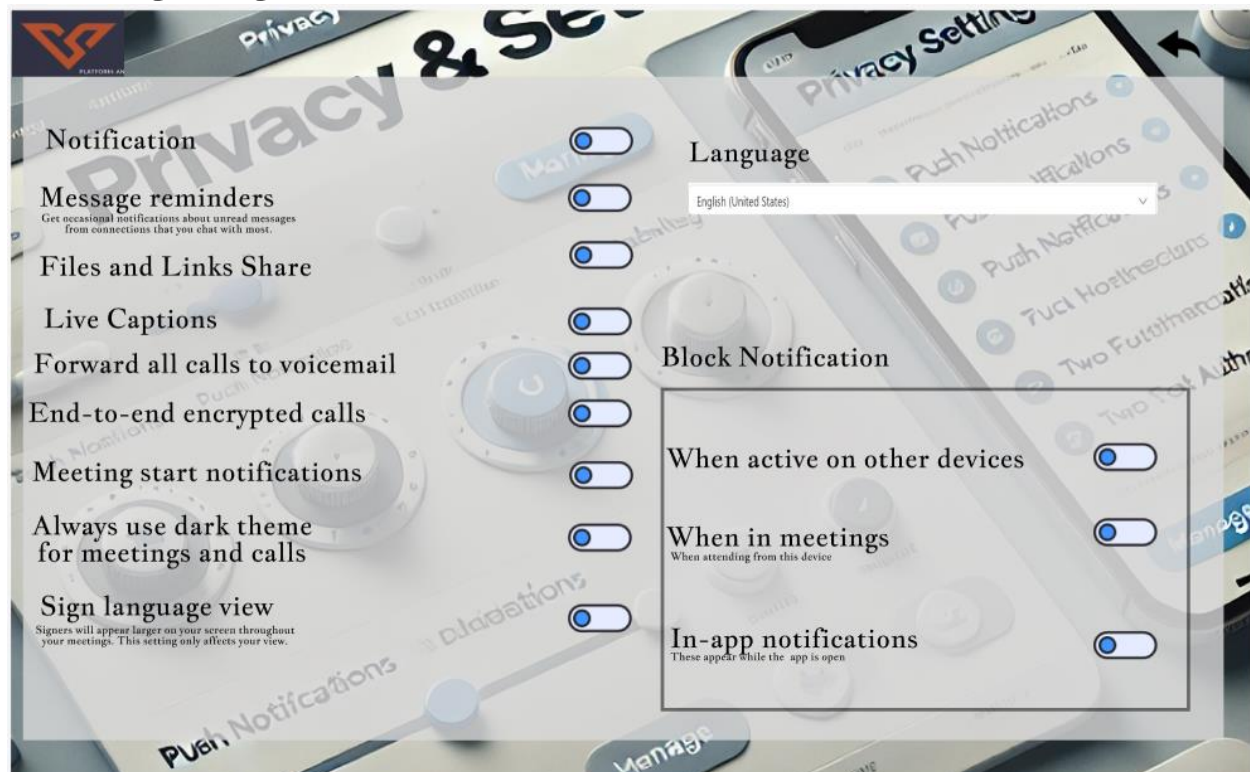
☒ Land

35. At the point of break-even sales, which one of

# Calendar page



# Setting Page



## Lab 2 Final: Tast case

|                                                                                                                |                                                                                                                                                      |                                                                                                                    |                |                    |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment |                                                                                                                                                      | Test Designed by: AFNAN BIN ISLAM NAHIN                                                                            |                |                    |
| Test Case ID: VC_01                                                                                            |                                                                                                                                                      | Test Designed date: 25-12-2024                                                                                     |                |                    |
| Test Priority (Low, Medium, High): High                                                                        |                                                                                                                                                      | Test Executed by: AFNAN                                                                                            |                |                    |
| Module Name: Login Session                                                                                     |                                                                                                                                                      | Test Execution date:01-1-2025                                                                                      |                |                    |
| Test Title: Verify login with valid email and password                                                         |                                                                                                                                                      |                                                                                                                    |                |                    |
| Description: Test the login functionality for valid credentials                                                |                                                                                                                                                      |                                                                                                                    |                |                    |
| Precondition (If any): User must have a valid account                                                          |                                                                                                                                                      |                                                                                                                    |                |                    |
| Test Steps                                                                                                     | Test Data                                                                                                                                            | Expected Results                                                                                                   | Actual Results | Status (Pass/Fail) |
| 1. Go to the website.<br>2. Enter email and password.<br>3. Click the "Continue" button.                       | URL: <a href="http://www.vclassroom.com">www.vclassroom.com</a><br>Email: <a href="mailto:afnan@gmail.com">afnan@gmail.com</a><br>Password: afnan123 | Website loads successfully.<br>User credentials are entered.<br>User is logged In and redirected to the dashboard. | As expected    | Pass               |
| Post Condition: User successfully logs in and accesses the dashboard.                                          |                                                                                                                                                      |                                                                                                                    |                |                    |

|                                                                                                                            |                                                                                                                                                                                         |                                                                                                                                                               |                |                    |
|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment             |                                                                                                                                                                                         | Test Designed by: AFNAN BIN ISLAM NAHIN                                                                                                                       |                |                    |
| Test Case ID: VC_12                                                                                                        |                                                                                                                                                                                         | Test Designed date: 25-12-2024                                                                                                                                |                |                    |
| Test Priority (Low, Medium, High): High                                                                                    |                                                                                                                                                                                         | Test Executed by: AFNAN                                                                                                                                       |                |                    |
| Module Name: User Registration (Sign-Up)                                                                                   |                                                                                                                                                                                         | Test Execution date:01-1-2025                                                                                                                                 |                |                    |
| Test Title: Test system validation for invalid data and duplicate accounts                                                 |                                                                                                                                                                                         |                                                                                                                                                               |                |                    |
| Description: Test the functionality of creating a new user account                                                         |                                                                                                                                                                                         |                                                                                                                                                               |                |                    |
| Precondition (If any): User must not have an existing account with the same email or username.                             |                                                                                                                                                                                         |                                                                                                                                                               |                |                    |
| Test Steps                                                                                                                 | Test Data                                                                                                                                                                               | Expected Results                                                                                                                                              | Actual Results | Status (Pass/Fail) |
| 1. Navigate to the Sign-Up page.<br>2. Enter valid user details.<br>3. Confirm password.<br>4. Click the "Sign Up" button. | URL: www.vclassroom.com/signup<br>Full Name: AFNAN NAHIN<br>Age : 18<br>Email: <a href="mailto:afnan@gmail.com">afnan@gmail.com</a><br>Password: afnan123<br>Confirm Password: afnan123 | Sign-Up page is displayed.<br>User details are entered successfully.<br>Password matches.<br>User is registered successfully and redirected to the dashboard. | As expected    | Pass               |
| Post Condition: A new user account is created, and the user is logged into the platform.                                   |                                                                                                                                                                                         |                                                                                                                                                               |                |                    |

|                                                                                                                |                                                       |                                                                                         |                                         |                    |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment |                                                       |                                                                                         | Test Designed by: AFNAN BIN ISLAM NAHIN |                    |
| Test Case ID: VC_13                                                                                            |                                                       |                                                                                         | Test Designed date: 25-12-2024          |                    |
| Test Priority (Low, Medium, High): High                                                                        |                                                       |                                                                                         | Test Executed by: AFNAN                 |                    |
| Module Name: Notifications                                                                                     |                                                       |                                                                                         | Test Execution date:01-1-2025           |                    |
| Test Title: Verify users receive push notifications                                                            |                                                       |                                                                                         |                                         |                    |
| Description: Test the push notification system for events and updates                                          |                                                       |                                                                                         |                                         |                    |
| Precondition (If any): Notifications must be enabled in user settings.                                         |                                                       |                                                                                         |                                         |                    |
| Test Steps                                                                                                     | Test Data                                             | Expected Results                                                                        | Actual Results                          | Status (Pass/Fail) |
| 1. Schedule a meeting.<br><br>2. Publish a new assignment.                                                     | Meeting ID: 50505<br><br>Assignment_Title: Homework 1 | Notification for scheduled meeting is sent.<br><br>Notification is sent to your friend. | As expected                             | Pass               |
| Post Condition: Notifications are sent and received as expected.                                               |                                                       |                                                                                         |                                         |                    |

|                                                                                                                |                   |                                                                                              |                |                    |
|----------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------|----------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment |                   | Test Designed by: AMAN ULLAH                                                                 |                |                    |
| Test Case ID: VC_20                                                                                            |                   | Test Designed date: 27-12-2024                                                               |                |                    |
| Test Priority (Low, Medium, High): Medium                                                                      |                   | Test Executed by: AMAN                                                                       |                |                    |
| Module Name: Attendance Tracking                                                                               |                   | Test Execution date:02-1-2025                                                                |                |                    |
| Test Title: Verify attendance tracking during a session                                                        |                   |                                                                                              |                |                    |
| Description: Test automatic attendance marking for participants                                                |                   |                                                                                              |                |                    |
| Precondition (If any): Attendance tracking is enabled for the session.                                         |                   |                                                                                              |                |                    |
| Test Steps                                                                                                     | Test Data         | Expected Results                                                                             | Actual Results | Status (Pass/Fail) |
| 1. Start a live session.<br>2. End the session.                                                                | Meeting ID: 50505 | Attendance is recorded for all participants who join.<br><br>Attendance report is generated. | As expected    | Pass               |
| Post Condition: Attendance details are saved and visible in the reports.                                       |                   |                                                                                              |                |                    |

|                                                                                                                |                   |                                                                                               |                |                    |
|----------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------|----------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment |                   | Test Designed by: AMAN ULLAH                                                                  |                |                    |
| Test Case ID: VC_25                                                                                            |                   | Test Designed date: 27-12-2024                                                                |                |                    |
| Test Priority (Low, Medium, High): Low                                                                         |                   | Test Executed by: AMAN                                                                        |                |                    |
| Module Name: Session Recording                                                                                 |                   | Test Execution date:02-1-2025                                                                 |                |                    |
| Test Title: Verify session recording functionality                                                             |                   |                                                                                               |                |                    |
| Description: Test recording and playback of live sessions.                                                     |                   |                                                                                               |                |                    |
| Precondition (If any): Recording option must be enabled                                                        |                   |                                                                                               |                |                    |
| Test Steps                                                                                                     | Test Data         | Expected Results                                                                              | Actual Results | Status (Pass/Fail) |
| 1. Start recording during session.<br>2. Stop recording.                                                       | Meeting ID: 70707 | Session recording starts successfully.<br><br>Recording is saved and accessible for playback. | As expected    | Pass               |
| Post Condition: Recording is stored and available for review.                                                  |                   |                                                                                               |                |                    |

|                                                                                                                |                                             |                                                                                                   |                |                    |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------|----------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment |                                             | Test Designed by: HRITTIK BHATTACHARJEE                                                           |                |                    |
| Test Case ID: VC_26                                                                                            |                                             | Test Designed date: 28-12-2024                                                                    |                |                    |
| Test Priority (Low, Medium, High): Medium                                                                      |                                             | Test Executed by: HRITTIK                                                                         |                |                    |
| Module Name: Screen Sharing                                                                                    |                                             | Test Execution date:03-1-2025                                                                     |                |                    |
| Test Title: Verify instructor can share the screen                                                             |                                             |                                                                                                   |                |                    |
| Description: Test screen-sharing functionality during a session                                                |                                             |                                                                                                   |                |                    |
| Precondition (If any): The instructor must be in a live session                                                |                                             |                                                                                                   |                |                    |
| Test Steps                                                                                                     | Test Data                                   | Expected Results                                                                                  | Actual Results | Status (Pass/Fail) |
| 1. Start a live session.<br>2. Click on "Share Screen."                                                        | Meeting ID: 70707<br><br>No additional data | Live session starts successfully.<br><br>The screen is successfully shared with all participants. | As expected    | Pass               |
| Post Condition: Participants can view the instructor's screen without interruptions.                           |                                             |                                                                                                   |                |                    |

|                                                                                                                |                   |                                                                                                                    |                                         |                    |
|----------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment |                   |                                                                                                                    | Test Designed by: HRITTIK BHATTACHARJEE |                    |
| Test Case ID: VC_27                                                                                            |                   |                                                                                                                    | Test Designed date: 28-12-2024          |                    |
| Test Priority (Low, Medium, High): High                                                                        |                   |                                                                                                                    | Test Executed by: HRITTIK               |                    |
| Module Name: Join Meeting                                                                                      |                   |                                                                                                                    | Test Execution date:03-1-2025           |                    |
| Test Title: Verify user can join a live session                                                                |                   |                                                                                                                    |                                         |                    |
| Description: Test the functionality of joining a live session.                                                 |                   |                                                                                                                    |                                         |                    |
| Precondition (If any): The user must have a valid account and a scheduled session.                             |                   |                                                                                                                    |                                         |                    |
| Test Steps                                                                                                     | Test Data         | Expected Results                                                                                                   | Actual Results                          | Status (Pass/Fail) |
| 1. Log in to the platform.<br>2. Navigate to the session.<br>3. Click "Join Meeting."                          | Meeting ID: 70707 | The user successfully logs in.<br><br>The session details are visible.<br><br>User successfully joins the session. | As expected                             | Pass               |
| Post Condition: User is connected to the session, and audio/video functionality works seamlessly.              |                   |                                                                                                                    |                                         |                    |

| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment                          |                                                                                                                                                                                                                                                                                                    | Test Designed by: MD.HACHIBUR RAHMAN                                                                                                                                                                  |                |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| Test Case ID: VC_28                                                                                                                     |                                                                                                                                                                                                                                                                                                    | Test Designed date: 25-12-2024                                                                                                                                                                        |                |                    |
| Test Priority (Low, Medium, High): High                                                                                                 |                                                                                                                                                                                                                                                                                                    | Test Executed by: HACHIBUR                                                                                                                                                                            |                |                    |
| Module Name: Create Meeting                                                                                                             |                                                                                                                                                                                                                                                                                                    | Test Execution date:01-1-2025                                                                                                                                                                         |                |                    |
| Test Title: Verify a meeting can be created with valid details                                                                          |                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                       |                |                    |
| Description: Test the functionality of scheduling a new meeting.                                                                        |                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                       |                |                    |
| Precondition (If any): The user must be logged in to the platform.                                                                      |                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                       |                |                    |
| Test Steps                                                                                                                              | Test Data                                                                                                                                                                                                                                                                                          | Expected Results                                                                                                                                                                                      | Actual Results | Status (Pass/Fail) |
| 1. Navigate to the "Create Meeting" page.<br>2. Enter meeting details.<br>3. Click the "Create Meeting" button.<br>4. Add participants. | URL: <a href="http://www.vclassroom.com/create-meeting">www.vclassroom.com/create-meeting</a><br><br>Participant_Emails: user1@example.com, <a href="mailto:user2@example.com">user2@example.com</a><br><br>Meeting Title: Team_Discussion<br>Date:31/12/2024<br>Time:10:00am<br>Duration: 60 mins | Create Meeting page is displayed.<br><br>Meeting details are entered successfully.<br><br>Participants are added to the meeting.<br><br>Meeting is successfully created, and meeting ID is generated. | As expected    | Pass               |
| Post Condition: A new meeting is created, and participants receive an invitation with the meeting details (ID and passcode).            |                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                       |                |                    |

|                                                                                                                |                                |                                                                                                                           |                                      |                    |
|----------------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment |                                |                                                                                                                           | Test Designed by: MD.HACHIBUR RAHMAN |                    |
| Test Case ID: VC_30                                                                                            |                                |                                                                                                                           | Test Designed date: 25-12-2024       |                    |
| Test Priority (Low, Medium, High): High                                                                        |                                |                                                                                                                           | Test Executed by: HACHIBUR           |                    |
| Module Name: Chat Feature                                                                                      |                                |                                                                                                                           | Test Execution date:01-1-2025        |                    |
| Test Title: Verify sending a chat message                                                                      |                                |                                                                                                                           |                                      |                    |
| Description: Test the ability of a user to send a chat message in a meeting.                                   |                                |                                                                                                                           |                                      |                    |
| Precondition (If any): User must be logged in and in the chat window.                                          |                                |                                                                                                                           |                                      |                    |
| Test Steps                                                                                                     | Test Data                      | Expected Results                                                                                                          | Actual Results                       | Status (Pass/Fail) |
| 1. Navigate to the chat window.<br>2. Leave the message input blank.<br>3. Click the "Send" button.            | Message:<br>"Hello, everyone!" | Chat window is displayed.<br><br>The message is visible in the input box.<br><br>Message is sent and appears in the chat. | As expected                          | Pass               |
| Post Condition: Messages from other participants are displayed in the chat window in real-time.                |                                |                                                                                                                           |                                      |                    |

|                                                                                                                |  |                        |                                                                                      |                |                    |
|----------------------------------------------------------------------------------------------------------------|--|------------------------|--------------------------------------------------------------------------------------|----------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment |  |                        | Test Designed by: AFNAN BIN ISLAM NAHIN                                              |                |                    |
| Test Case ID: VC_32                                                                                            |  |                        | Test Designed date: 25-12-2024                                                       |                |                    |
| Test Priority (Low, Medium, High): High                                                                        |  |                        | Test Executed by: AFNAN                                                              |                |                    |
| Module Name: Call Feature                                                                                      |  |                        | Test Execution date:01-1-2025                                                        |                |                    |
| Test Title: Verify starting a call                                                                             |  |                        |                                                                                      |                |                    |
| Description: Test if a user can successfully start a video or audio call.                                      |  |                        |                                                                                      |                |                    |
| Precondition (If any): User must be logged in and part of a virtual classroom.                                 |  |                        |                                                                                      |                |                    |
| Test Steps                                                                                                     |  | Test Data              | Expected Results                                                                     | Actual Results | Status (Pass/Fail) |
| 1. Navigate to the "Call" section.<br>2. Click the "Start Call" button.                                        |  | Call Type: Video/Audio | Call interface is displayed.<br><br>Call initiates, and "Call in Progress" is shown. | As expected    | Pass               |
| Post Condition: The user is in an active call session.                                                         |  |                        |                                                                                      |                |                    |

|                                                                                                                                                                                 |                                                                                                                |                                                                                                                                                                                                                                                                    |                                         |                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment                                                                  |                                                                                                                |                                                                                                                                                                                                                                                                    | Test Designed by: AFNAN BIN ISLAM NAHIN |                    |
| Test Case ID: VC_33                                                                                                                                                             |                                                                                                                |                                                                                                                                                                                                                                                                    | Test Designed date: 25-12-2024          |                    |
| Test Priority (Low, Medium, High): Medium                                                                                                                                       |                                                                                                                |                                                                                                                                                                                                                                                                    | Test Executed by: AFNAN                 |                    |
| Module Name: Create Assignment                                                                                                                                                  |                                                                                                                |                                                                                                                                                                                                                                                                    | Test Execution date:01-1-2025           |                    |
| Test Title: Verify creating an assignment                                                                                                                                       |                                                                                                                |                                                                                                                                                                                                                                                                    |                                         |                    |
| Description: Test if a user can successfully create an assignment.                                                                                                              |                                                                                                                |                                                                                                                                                                                                                                                                    |                                         |                    |
| Precondition (If any): User must be logged in and have access to the assignment creation module.                                                                                |                                                                                                                |                                                                                                                                                                                                                                                                    |                                         |                    |
| Test Steps                                                                                                                                                                      | Test Data                                                                                                      | Expected Results                                                                                                                                                                                                                                                   | Actual Results                          | Status (Pass/Fail) |
| 1. Navigate to the "Assignment" section.<br>2. Click on the "Create Assignment" button.<br>3. Enter assignment details.<br>4. Attach resources .<br>5. Click on "Share" button. | Title: Math Assignment<br>Description: Solve problems 1-10<br>Due Date: 2025-01-10<br><br>File: "Problems.pdf" | Assignment creation page is displayed.<br><br>Assignment creation form is displayed.<br><br>Assignment details are entered successfully.<br><br>File is successfully uploaded and attached to the assignment.<br><br>Assignment is successfully created and share. | As expected                             | Pass               |
| Post Condition: The assignment is visible to those the user share it with .                                                                                                     |                                                                                                                |                                                                                                                                                                                                                                                                    |                                         |                    |

|                                                                                                                                                                 |                                                                                                                                                                      |                                                                                                                                                                                                                                |                                         |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment                                                  |                                                                                                                                                                      |                                                                                                                                                                                                                                | Test Designed by: AFNAN BIN ISLAM NAHIN |                    |
| Test Case ID: VC_50                                                                                                                                             |                                                                                                                                                                      |                                                                                                                                                                                                                                | Test Designed date: 25-12-2024          |                    |
| Test Priority (Low, Medium, High): High                                                                                                                         |                                                                                                                                                                      |                                                                                                                                                                                                                                | Test Executed by: AFNAN                 |                    |
| Module Name: Create Exam                                                                                                                                        |                                                                                                                                                                      |                                                                                                                                                                                                                                | Test Execution date:01-1-2025           |                    |
| Test Title: Verify creating an exam                                                                                                                             |                                                                                                                                                                      |                                                                                                                                                                                                                                |                                         |                    |
| Description: Test if a user can successfully create an exam.                                                                                                    |                                                                                                                                                                      |                                                                                                                                                                                                                                |                                         |                    |
| Precondition (If any): User must be logged in and have access to the exam creation module.                                                                      |                                                                                                                                                                      |                                                                                                                                                                                                                                |                                         |                    |
| Test Steps                                                                                                                                                      | Test Data                                                                                                                                                            | Expected Results                                                                                                                                                                                                               | Actual Results                          | Status (Pass/Fail) |
| 1. Navigate to the "Assignment" section.<br>2. Click on the "Create Exam" button.<br>3. Enter exam details.<br>4. Add questions.<br>5. Click on "Share" button. | Title: Final Math Exam<br>Description:<br>Covers Chapters 1-5<br>Duration: 90 minutes<br>Time: 10 AM<br><br>Questions:<br>Multiple Choice, Short Answers, True/False | Exam creation page is displayed.<br><br>Exam creation form is displayed.<br><br>Exam details are entered successfully.<br><br>Questions are added to the exam successfully.<br><br>Exam is successfully created and scheduled. | As expected                             | Pass               |
| Post Condition: The exam is visible to those the user share it with and is scheduled for the specified date and time.                                           |                                                                                                                                                                      |                                                                                                                                                                                                                                |                                         |                    |

|                                                                                                                |                             |                                                                                                                                                               |                                    |                    |
|----------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment |                             |                                                                                                                                                               | Test Designed by: WALID BIN KHALID |                    |
| Test Case ID: VC_51                                                                                            |                             |                                                                                                                                                               | Test Designed date: 25-12-2024     |                    |
| Test Priority (Low, Medium, High): High                                                                        |                             |                                                                                                                                                               | Test Executed by: WALID            |                    |
| Module Name: Attend Exam                                                                                       |                             |                                                                                                                                                               | Test Execution date:01-1-2025      |                    |
| Test Title: Verify the ability to start an exam                                                                |                             |                                                                                                                                                               |                                    |                    |
| Description: Test if a user can successfully start an exam                                                     |                             |                                                                                                                                                               |                                    |                    |
| Precondition (If any): A published exam must exist, and the user should be enrolled in the respective course.  |                             |                                                                                                                                                               |                                    |                    |
| Test Steps                                                                                                     | Test Data                   | Expected Results                                                                                                                                              | Actual Results                     | Status (Pass/Fail) |
| 1.Navigate to the "Assienment" section.<br><br>2.Click the "Attend Exam" button.<br><br>3. Select an exam.     | Exam Title: Final Math Exam | List of available exams is displayed.<br><br>Exam details and "view Exam" button are displayed.<br><br>The exam interface is displayed, and the timer starts. | As expected                        | Pass               |
| Post Condition: The user successfully starts the exam and can see the questions along with the timer.          |                             |                                                                                                                                                               |                                    |                    |

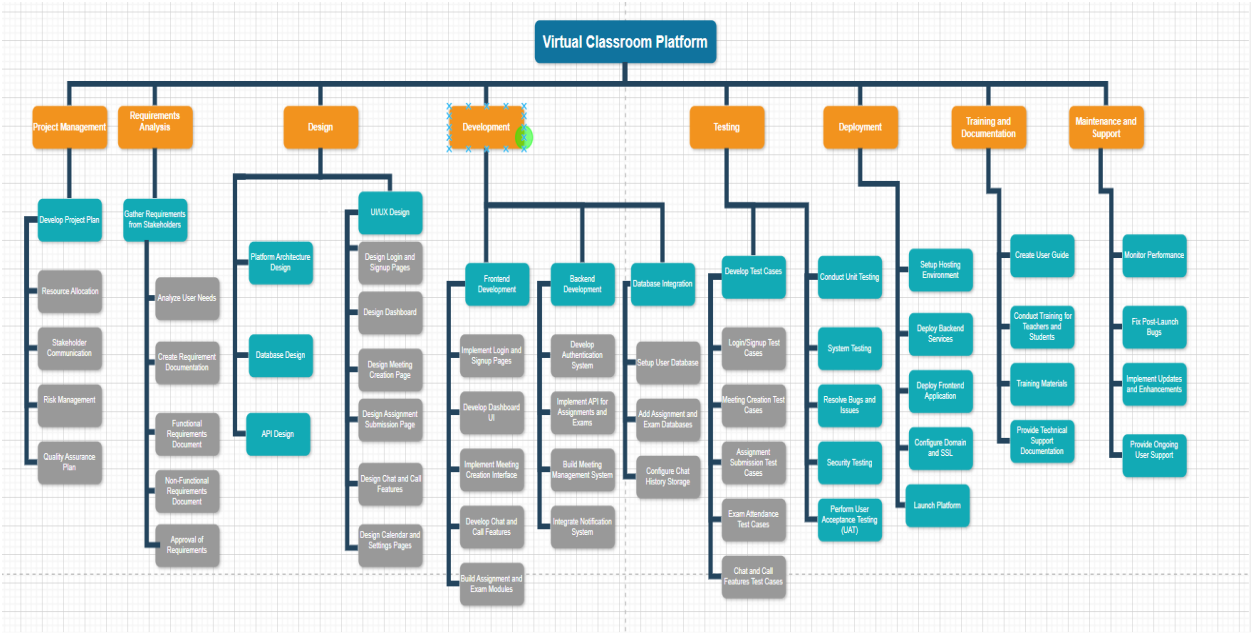
|                                                                                                                                                                                         |                                                                             |                                                                                                                                                                         |                |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment                                                                          |                                                                             | Test Designed by: AFNAN BIN ISLAM NAHIN                                                                                                                                 |                |                    |
| Test Case ID: VC_52                                                                                                                                                                     |                                                                             | Test Designed date: 25-12-2024                                                                                                                                          |                |                    |
| Test Priority (Low, Medium, High): High                                                                                                                                                 |                                                                             | Test Executed by: AFNAN                                                                                                                                                 |                |                    |
| Module Name: Submit Assignment                                                                                                                                                          |                                                                             | Test Execution date:01-1-2025                                                                                                                                           |                |                    |
| Test Title: Verify successful assignment submission                                                                                                                                     |                                                                             |                                                                                                                                                                         |                |                    |
| Description: Test if a user can upload and submit an assignment successfully.                                                                                                           |                                                                             |                                                                                                                                                                         |                |                    |
| Precondition (If any): The assignment must be published by a user, and to those the user share it should be enrolled in the respective course.                                          |                                                                             |                                                                                                                                                                         |                |                    |
| Test Steps                                                                                                                                                                              | Test Data                                                                   | Expected Results                                                                                                                                                        | Actual Results | Status (Pass/Fail) |
| 1. Navigate to the "Assignments" section.<br><br>2.Click on "Submit Assignment."<br><br>3. Select the assignment.<br><br>4. Click on "Attach Assignment."<br><br>5. Click on "Turn in." | Exam Title: Final Math Exam<br><br><br><br><br><br>File: math_homework1.pdf | . Student dashboard is displayed.<br><br>List of available assignments is displayed.<br><br>Assignment details page is displayed.<br><br>File is uploaded successfully. | As expected    | Pass               |
| Post Condition: The assignment is submitted, and the submission timestamp is recorded in the system.                                                                                    |                                                                             |                                                                                                                                                                         |                |                    |

|                                                                                                                |                                                                                              |                                                                                                    |                                    |                    |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment |                                                                                              |                                                                                                    | Test Designed by: WALID BIN KHALID |                    |
| Test Case ID: VC_53                                                                                            |                                                                                              |                                                                                                    | Test Designed date: 25-12-2024     |                    |
| Test Priority (Low, Medium, High): High                                                                        |                                                                                              |                                                                                                    | Test Executed by: WALID            |                    |
| Module Name: Calendar                                                                                          |                                                                                              |                                                                                                    | Test Execution date:01-1-2025      |                    |
| Test Title: Verify event creation in the calendar                                                              |                                                                                              |                                                                                                    |                                    |                    |
| Description: Test if users can create and add events to the calendar.                                          |                                                                                              |                                                                                                    |                                    |                    |
| Precondition (If any): User must be logged in and have access to the calendar module.                          |                                                                                              |                                                                                                    |                                    |                    |
| Test Steps                                                                                                     | Test Data                                                                                    | Expected Results                                                                                   | Actual Results                     | Status (Pass/Fail) |
| 1. Log in to the platform.<br>2. Navigate to the "Calendar" section.<br>3. Click on "Add Event."               | Event Title: "Math Quiz"<br>Start Date: 2025-01-10<br>End Date: 2025-01-10<br>Time: 10:00 AM | . Calendar interface is displayed.<br><br>Event is added to the calendar with the correct details. | As expected                        | Pass               |
| Post Condition: The event is saved and displayed on the calendar.                                              |                                                                                              |                                                                                                    |                                    |                    |

|                                                                                                                                                                                                                                                                       |                       |                                                                                                                                                                                                                                        |                                         |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------|
| Project Name: Virtual Classroom Platform: An Interactive, Accessible, and Engaging Online Learning Environment                                                                                                                                                        |                       |                                                                                                                                                                                                                                        | Test Designed by: AFNAN BIN ISLAM NAHIN |                    |
| Test Case ID: ST_01                                                                                                                                                                                                                                                   |                       |                                                                                                                                                                                                                                        | Test Designed date: 25-12-2024          |                    |
| Test Priority (Low, Medium, High): High                                                                                                                                                                                                                               |                       |                                                                                                                                                                                                                                        | Test Executed by: AFNAN                 |                    |
| Module Name: Settings Page                                                                                                                                                                                                                                            |                       |                                                                                                                                                                                                                                        | Test Execution date:01-1-2025           |                    |
| Test Title: Verify enabling and disabling notification preferences                                                                                                                                                                                                    |                       |                                                                                                                                                                                                                                        |                                         |                    |
| Description: Test the functionality of toggling notification preferences in the Settings Page.                                                                                                                                                                        |                       |                                                                                                                                                                                                                                        |                                         |                    |
| Precondition (If any): <ul style="list-style-type: none"><li>• The updated settings should persist after refreshing the page.</li><li>• Notifications should function according to the enabled or disabled preferences.</li></ul>                                     |                       |                                                                                                                                                                                                                                        |                                         |                    |
| Test Steps                                                                                                                                                                                                                                                            | Test Data             | Expected Results                                                                                                                                                                                                                       | Actual Results                          | Status (Pass/Fail) |
| 1.Navigate to the Settings Page.<br><br>2.Toggle "Message Reminders" to <b>ON</b> .<br><br>3Toggle "Files and Links Share" to <b>OFF</b> .<br><br>4. Toggle "End-to-end encrypted calls" to <b>ON</b> .<br><br>5. Select a different language from the dropdown menu. | Select "English (UK)" | Settings page loads successfully.<br><br>The toggle should switch to <b>ON</b> .<br><br>The toggle should switch to <b>OFF</b> .<br><br>The toggle should switch to <b>ON</b> .<br><br>Language preference should update successfully. | As expected                             | Pass               |
| Post Condition: <ul style="list-style-type: none"><li>• The updated settings should persist after refreshing the page.</li><li>• Notifications should function according to the enabled or disabled preferences.</li></ul>                                            |                       |                                                                                                                                                                                                                                        |                                         |                    |

Lab 3 Final:

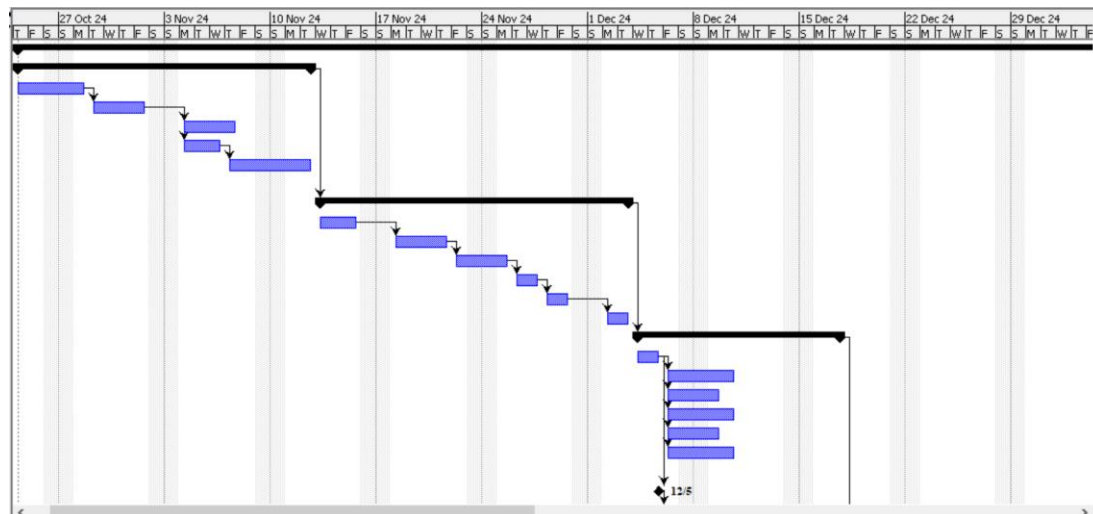
Work Breakdown Structure



## Gantt Chart

|    |   | Name                                  | Duration | Start            | Finish           | Predecessors |
|----|---|---------------------------------------|----------|------------------|------------------|--------------|
| 1  | ✖ | Virtual Classroom Platform            | 87 days  | 10/24/24 8:00 AM | 2/21/25 5:00 PM  |              |
| 2  |   | Project Management                    | 14 days  | 10/24/24 8:00 AM | 11/12/24 5:00 PM |              |
| 3  |   | Develop Project Plan                  | 3 days   | 10/24/24 8:00 AM | 10/28/24 5:00 PM |              |
| 4  |   | Resource Allocation                   | 4 days   | 10/29/24 8:00 AM | 11/1/24 5:00 PM  | 3            |
| 5  |   | Stakeholder Communication             | 4 days   | 11/4/24 8:00 AM  | 11/7/24 5:00 PM  | 4            |
| 6  |   | Risk Management                       | 3 days   | 11/4/24 8:00 AM  | 11/6/24 5:00 PM  | 4            |
| 7  |   | Quality Assurance Plan                | 4 days   | 11/7/24 8:00 AM  | 11/12/24 5:00 PM | 6            |
| 8  |   | Requirements Analysis                 | 15 days  | 11/13/24 8:00 AM | 12/3/24 5:00 PM  | 2            |
| 9  |   | Gather Requirements from Stakeholders | 3 days   | 11/13/24 8:00 AM | 11/15/24 5:00 PM |              |
| 10 |   | Analyze User Needs                    | 4 days   | 11/18/24 8:00 AM | 11/21/24 5:00 PM | 9            |
| 11 |   | Create Requirement Documents          | 2 days   | 11/22/24 8:00 AM | 11/25/24 5:00 PM | 10           |
| 12 |   | Functional Requirements               | 2 days   | 11/26/24 8:00 AM | 11/27/24 5:00 PM | 11           |
| 13 |   | Non-Functional Requirements           | 2 days   | 11/28/24 8:00 AM | 11/29/24 5:00 PM | 12           |
| 14 |   | Approval of Requirements              | 2 days   | 12/2/24 8:00 AM  | 12/3/24 5:00 PM  | 13           |
| 15 |   | Design                                | 10 days  | 12/4/24 8:00 AM  | 12/17/24 5:00 PM | 8            |
| 16 |   | Platform Architecture Design          | 2 days   | 12/4/24 8:00 AM  | 12/5/24 5:00 PM  |              |
| 17 |   | Class diagram                         | 3 days   | 12/6/24 8:00 AM  | 12/10/24 5:00 PM | 16           |
| 18 |   | ER-diagram                            | 2 days   | 12/6/24 8:00 AM  | 12/9/24 5:00 PM  | 16           |
| 19 |   | sequence diagram                      | 3 days   | 12/6/24 8:00 AM  | 12/10/24 5:00 PM | 16           |
| 20 |   | use case diagram                      | 2 days   | 12/6/24 8:00 AM  | 12/9/24 5:00 PM  | 16           |
| 21 |   | activity diagram                      | 3 days   | 12/6/24 8:00 AM  | 12/10/24 5:00 PM | 16           |
| 22 |   | Database Design                       | 0 days   | 12/5/24 5:00 PM  | 12/5/24 5:00 PM  | 16           |

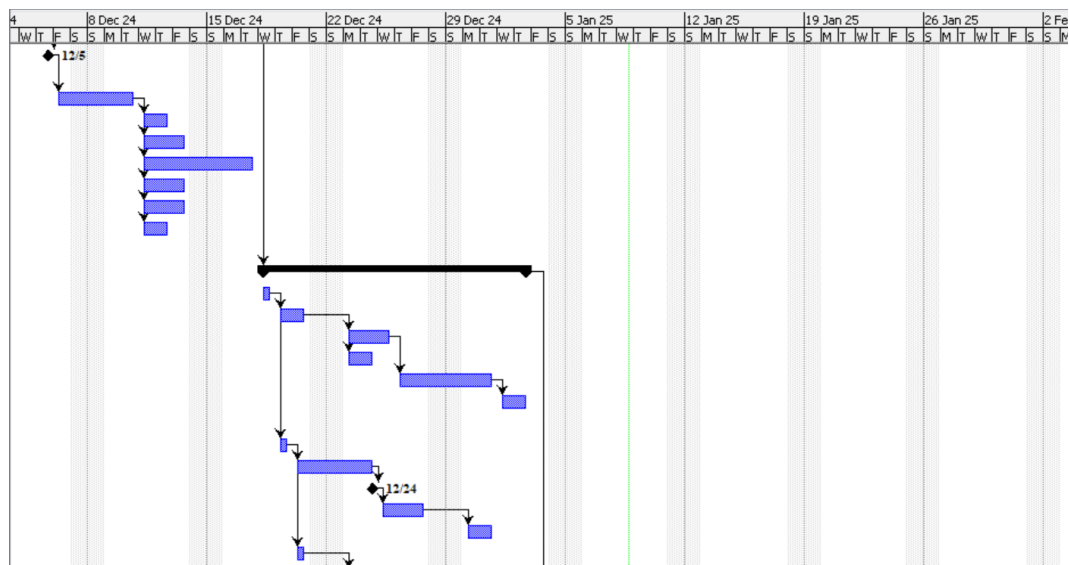
Fig:1



(this timeline chart is from fig-1 line(1-22))

|    | 🔍 | Name                       | Duration | Start            | Finish           | Pr |
|----|---|----------------------------|----------|------------------|------------------|----|
| 23 |   | API Design                 | 0 days   | 12/5/24 5:00 PM  | 12/5/24 5:00 PM  | 22 |
| 24 |   | UI/UX Design               | 3 days   | 12/6/24 8:00 AM  | 12/10/24 5:00 PM | 23 |
| 25 |   | Design Login and Signup    | 2 days   | 12/11/24 8:00 AM | 12/12/24 5:00 PM | 24 |
| 26 |   | Design Dashboard           | 3 days   | 12/11/24 8:00 AM | 12/13/24 5:00 PM | 24 |
| 27 |   | Design Meeting Creation    | 5 days   | 12/11/24 8:00 AM | 12/17/24 5:00 PM | 24 |
| 28 |   | Design Assignment Submi    | 3 days   | 12/11/24 8:00 AM | 12/13/24 5:00 PM | 24 |
| 29 |   | Design Chat and Call Fea   | 3 days   | 12/11/24 8:00 AM | 12/13/24 5:00 PM | 24 |
| 30 |   | Design Calendar and Sett   | 2 days   | 12/11/24 8:00 AM | 12/12/24 5:00 PM | 24 |
| 31 | 📅 | Development                | 12 days  | 12/18/24 8:00 AM | 1/2/25 5:00 PM   | 15 |
| 32 |   | Frontend Development       | 1 day    | 12/18/24 8:00 AM | 12/18/24 5:00 PM |    |
| 33 |   | Implement Login and Sign   | 2 days   | 12/19/24 8:00 AM | 12/20/24 5:00 PM | 32 |
| 34 |   | Develop Dashboard UI       | 3 days   | 12/23/24 8:00 AM | 12/25/24 5:00 PM | 33 |
| 35 |   | Implement Meeting Creat    | 2 days   | 12/23/24 8:00 AM | 12/24/24 5:00 PM | 33 |
| 36 |   | Develop Chat and Call Fe   | 4 days   | 12/26/24 8:00 AM | 12/31/24 5:00 PM | 34 |
| 37 |   | Build Assignment and Exa   | 2 days   | 1/1/25 8:00 AM   | 1/2/25 5:00 PM   | 36 |
| 38 |   | Backend Development        | 1 day    | 12/19/24 8:00 AM | 12/19/24 5:00 PM | 32 |
| 39 |   | Develop Authentication S   | 3 days   | 12/20/24 8:00 AM | 12/24/24 5:00 PM | 38 |
| 40 |   | Implement API for Assign   | 0 days   | 12/24/24 5:00 PM | 12/24/24 5:00 PM | 39 |
| 41 |   | Build Meeting Managemer    | 3 days   | 12/25/24 8:00 AM | 12/27/24 5:00 PM | 40 |
| 42 |   | Integrate Notification Sys | 2 days   | 12/30/24 8:00 AM | 12/31/24 5:00 PM | 41 |
| 43 |   | Database Integration       | 1 day    | 12/20/24 8:00 AM | 12/20/24 5:00 PM | 38 |

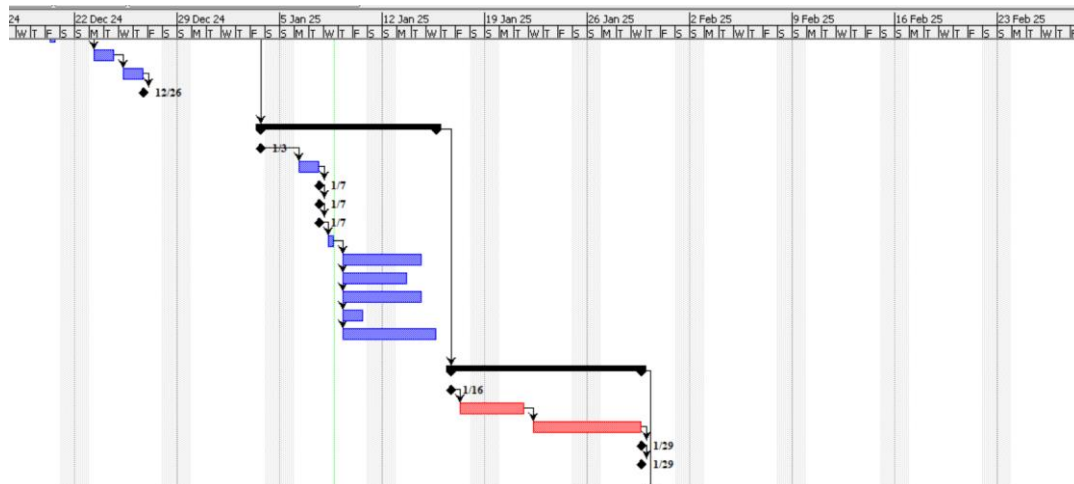
Fig:2



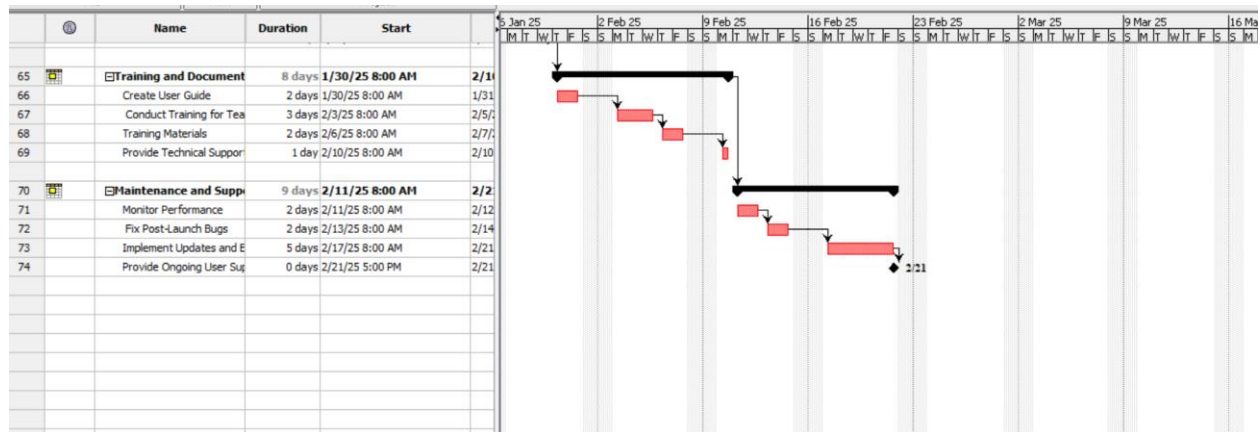
(this timeline chart is from fig-2 line(23-43))

|    | Ⓜ | Name                      | Duration | Start            | Finish           | Predecessors |
|----|---|---------------------------|----------|------------------|------------------|--------------|
| 44 |   | Setup User Database       | 2 days   | 12/23/24 8:00 AM | 12/24/24 5:00 PM | 43           |
| 45 |   | Add Assignment and Exam   | 2 days   | 12/25/24 8:00 AM | 12/26/24 5:00 PM | 44           |
| 46 |   | Configure Chat History S  | 0 days   | 12/26/24 5:00 PM | 12/26/24 5:00 PM | 45           |
| 47 | 📅 | Testing                   | 8 days   | 1/3/25 5:00 PM   | 1/15/25 5:00 PM  | 31           |
| 48 |   | Conduct Unit Testing      | 0 days   | 1/3/25 5:00 PM   | 1/3/25 5:00 PM   |              |
| 49 |   | System Testing            | 2 days   | 1/6/25 8:00 AM   | 1/7/25 5:00 PM   | 48           |
| 50 |   | Resolve Bugs and Issues   | 0 days   | 1/7/25 5:00 PM   | 1/7/25 5:00 PM   | 49           |
| 51 |   | Security Testing          | 0 days   | 1/7/25 5:00 PM   | 1/7/25 5:00 PM   | 50           |
| 52 |   | Perform User Acceptance   | 0 days   | 1/7/25 5:00 PM   | 1/7/25 5:00 PM   | 51           |
| 53 |   | Develop Test Cases        | 1 day    | 1/8/25 8:00 AM   | 1/8/25 5:00 PM   | 52           |
| 54 |   | Login/Signup Test Cases   | 4 days   | 1/9/25 8:00 AM   | 1/14/25 5:00 PM  | 53           |
| 55 |   | Meeting Creation Test Ca  | 3 days   | 1/9/25 8:00 AM   | 1/13/25 5:00 PM  | 53           |
| 56 |   | Assignment Submission Te  | 4 days   | 1/9/25 8:00 AM   | 1/14/25 5:00 PM  | 53           |
| 57 |   | Exam Attendance Test Ca   | 2 days   | 1/9/25 8:00 AM   | 1/10/25 5:00 PM  | 53           |
| 58 | 📅 | Chat and Call Features Te | 5 days   | 1/9/25 8:00 AM   | 1/15/25 5:00 PM  | 53           |
| 59 | 📅 | Deployment                | 9 days   | 1/16/25 5:00 PM  | 1/29/25 5:00 PM  | 47           |
| 60 |   | Setup Hosting Environme   | 0 days   | 1/16/25 5:00 PM  | 1/16/25 5:00 PM  |              |
| 61 |   | Deploy Backend Services   | 3 days   | 1/17/25 8:00 AM  | 1/21/25 5:00 PM  | 60           |
| 62 |   | Deploy Frontend Applicat  | 6 days   | 1/22/25 8:00 AM  | 1/29/25 5:00 PM  | 61           |
| 63 |   | Configure Domain and SS   | 0 days   | 1/29/25 5:00 PM  | 1/29/25 5:00 PM  | 62           |
| 64 |   | Launch Platform           | 0 days   | 1/29/25 5:00 PM  | 1/29/25 5:00 PM  | 63           |

Fig:3



(this timeline chart is from fig-3 line(44-64))



(this timeline chart is from line(65-74))

## Lab 4 Final:

### Project Estimation (COMOMO)

For our Organic system project , let's take the number of lines of code to be,

SLOC =20000,

Effort Factor Coefficient = 2.4,

P= 1.05 and

T = 0.38.

We know ,

Effort = PM = Coefficient<sub><Effort Factor></sub>\*(SLOC/1000)<sup>P</sup>

Effort (PM) = 2.4 x (20000/1000)<sup>1.05</sup>

Effort (PM) =55.756

Development time = DM = 2.50\*(PM)<sup>T</sup>

= 2.50 x (55.756)<sup>0.38</sup>

= 11.5220 Months

Required number of people = ST = PM/DM

=55.756/ 11.5220

=4.839 ~ 5

## Human Resource Allocation

| Resource Name           | Role          | Responsibilities                                                                   | Skill Set Required                       | Allocation (Hours/Week) | Duration                |
|-------------------------|---------------|------------------------------------------------------------------------------------|------------------------------------------|-------------------------|-------------------------|
| Project Manager         | Manager       | Plan, monitor, and control project activities, ensure risk management, and QA      | Leadership, Communication, PM Tools      | 20                      | 01/01/2025 - 03/31/2025 |
| Business Analyst        | Analyst       | Gather requirements, interact with stakeholders, and document user needs           | Communication, Analytical, Documentation | 15                      | 01/01/2025 - 01/31/2025 |
| UI/UX Designer          | Designer      | Design platform layout, create interactive prototypes, and improve user experience | Figma, Adobe XD, Creativity              | 25                      | 02/01/2025 - 02/28/2025 |
| Frontend Developer      | Developer     | Build UI components, connect APIs, and ensure responsiveness                       | React.js, HTML/CSS, JavaScript           | 30                      | 03/01/2025 - 05/31/2025 |
| Backend Developer       | Developer     | Develop APIs, authentication, and logic for backend services                       | Node.js, Python, Database Management     | 30                      | 03/01/2025 - 05/31/2025 |
| Database Administrator  | Administrator | Design, implement, and optimize databases                                          | SQL, MongoDB, Backup/Recovery Skills     | 20                      | 03/01/2025-05/15/2025   |
| Quality Analyst         | Tester        | Create and execute test cases, conduct UAT, and debug                              | Test Automation, Problem-Solving         | 25                      | 05/01/2025 - 06/15/2025 |
| Deployment Engineer     | Engineer      | Setup hosting, deploy application, and manage configurations                       | AWS, Azure, CI/CD                        | 20                      | 06/01/2025 - 06/15/2025 |
| Trainer                 | Educator      | Conduct training for users and provide documentation                               | Communication, Training, Documentation   | 10                      | 06/15/2025 - 06/30/2025 |
| Technical Support Staff | Support       | Monitor platform, resolve user issues, and provide updates                         | Troubleshooting, Communication           | 10                      | 07/01/2025 - Ongoing    |

## Environment Resource Allocation

| Environment Type        | Purpose                                                       | Resources Required                                                                  | Allocated Duration      | Key Dependencies                   |
|-------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------|------------------------------------|
| Development Environment | For coding, debugging, and integrating components             | Workstations, IDEs (e.g., Visual Studio, IntelliJ), GitHub repository, test servers | January 2025 - May 2025 | Version Control, Developer Access  |
| Testing Environment     | For conducting unit, integration, and user acceptance testing | Test servers, automated testing tools (e.g., Selenium, JMeter), staging databases   | April 2025 - June 2025  | QA Team Access, Test Data          |
| Staging Environment     | Simulates production for final validation before deployment   | Staging servers, replica database, network configuration                            | May 2025 - June 2025    | Final Build, Deployment Approval   |
| Production Environment  | Live platform for end-users                                   | AWS EC2 instances, S3 storage, SSL certificate, Load balancers                      | June 2025 - Ongoing     | Scheduled Backup Configuration     |
| Backup Environment      | For data backups and disaster recovery                        | Cloud storage, Backup software (e.g., AWS Backup, Azure Backup)                     | March 2025 - Ongoing    | Trainer Access, Training Materials |
| Training Environment    | For user training and simulation                              | simulation<br>Training servers, replica platform instance, dummy user accounts      | June 2025               | Trainer Access, Training Materials |

## Re-usable Software Resource Allocation

| Resource Name         | Purpose                                 | Modules/Tasks Used                            | Licensing/Ownership | Allocated Duration        |
|-----------------------|-----------------------------------------|-----------------------------------------------|---------------------|---------------------------|
| Node.js               | Backend development and API creation    | Backend Development (Authentication, APIs)    | Open Source         | January 2025 - May 2025   |
| React.js              | Frontend development for user interface | Dashboard, Chat Interface, Meeting UI         | Open Source         | January 2025 - May 2025   |
| MySQL                 | Database management                     | User Data, Assignments, Exams                 | Open Source         | January 2025 - Ongoing    |
| Figma                 | UI/UX design prototyping                | prototyping Login, Dashboard, Calendar Design | Freemium            | January 2025 - March 2025 |
| AWS Cloud Services    | Hosting and storage                     | Production Environment                        | Pay-as-you-go       | May 2025 - Ongoing        |
| Jest                  | Unit testing for JavaScript code        | Frontend and Backend Unit Tests               | Open Source         | April 2025 - May 2025     |
| Bootstrap/Material UI | UI design framework                     | Frontend Components (Forms, Buttons)          | Open Source         | February 2025 - May 2025  |

Risk Analysis:

| Impact      |           | Very Low    | Low         | Medium      | High     | Very high |  |  |             |           |
|-------------|-----------|-------------|-------------|-------------|----------|-----------|--|--|-------------|-----------|
| Probability | Very high | Moderate    | Severe      | Severe      | Critical | Critical  |  |  |             |           |
|             | High      | Sustainable | Moderate    | Severe      | Critical | Critical  |  |  |             |           |
|             | Medium    | Sustainable | Moderate    | Moderate    | Severe   | Critical  |  |  | Probability | Impact    |
|             | Low       | Sustainable | Sustainable | Moderate    | Severe   | Critical  |  |  | Very high   | Very high |
|             | Very Low  | Sustainable | Sustainable | Sustainable | Moderate | Severe    |  |  | High        | High      |
|             |           |             |             |             |          |           |  |  | Medium      | Medium    |
|             |           |             |             |             |          |           |  |  | Low         | Low       |
|             |           |             |             |             |          |           |  |  | Very Low    | Very Low  |

| Unique id | Risk description                      | Caused by                  | Risk owner       | Probability | Impact    | Risk rating | Controls                                  | Controls owners | Residual probability | Residual Impact | Residual rating |
|-----------|---------------------------------------|----------------------------|------------------|-------------|-----------|-------------|-------------------------------------------|-----------------|----------------------|-----------------|-----------------|
|           |                                       |                            |                  |             |           |             |                                           |                 |                      |                 |                 |
|           | Bugs in critical functionalities      | Software defects           | Development Team | High        | Medium    | Moderate    | Robust testing process                    | QA Team         | Medium               | Medium          | Moderate        |
|           | Performance issues during peak usage  | System overload            | Development Team | High        | High      | Critical    | Load testing, scalable architecture       | DevOps Team     | Medium               | Medium          | Severe          |
|           | Scope creep                           | Changing user requirements | Project Manager  | Medium      | High      | Severe      | Clear scope, change management protocols  | Project Manager | Low                  | Medium          | Moderate        |
|           | Unauthorized access to sensitive data | Security vulnerabilities   | Security Team    | Medium      | Very High | Critical    | Data encryption, MFA, penetration testing | Security Team   | Low                  | High            | Severe          |
|           | Low adoption rate                     | User unfamiliarity         | Product Manager  | Medium      | Medium    | Moderate    | User training, intuitive design           | Training Team   | Low                  | Low             | Sustainable     |
|           | Budget overruns                       | Underestimated costs       | Product Manager  | Medium      | Medium    | Moderate    | Contingency funds, expense tracking       | Finance Team    | Low                  | Medium          | Moderate        |
|           |                                       |                            |                  |             |           |             |                                           |                 |                      |                 |                 |
|           | Delays in resource allocation         | Resource unavailability    | HR Team          | Medium      | High      | Severe      | Flexible timelines, resource backups      | Project Manager | Low                  | Medium          | Moderate        |
|           | Data loss due to system crashes       | Hardware/software failures | IT Team          | Low         | High      | Moderate    | Regular backups, recovery systems         | IT Team         | Low                  | Low             | Sustainable     |
|           | Negative feedback on usability        | Poor user experience       | Design Team      | Medium      | Medium    | Moderate    | Beta testing, incorporate user feedback   | UX/UI Team      | Low                  | Medium          | Moderate        |
|           | Post-launch system downtime           | Maintenance errors         | DevOps Team      | Low         | Medium    | Sustainable | Monitoring tools, dedicated support team  | DevOps Team     | Very Low             | Medium          | Sustainable     |